

Project Synopsis

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Title

Autonomous Cyber Defense System using AI for Real-Time Threat Hunting in Cloud ERP (Prototype)

Problem Statement (200 words)

With the increasing adoption of cloud-integrated ERP systems in modern organizations, ensuring robust cybersecurity has become a significant challenge. Traditional security systems rely heavily on manual monitoring and predefined rules, which are often ineffective against evolving and sophisticated cyber threats. These approaches fail to provide real-time detection and proactive defense mechanisms.

Cloud environments generate a large volume of logs through services such as AWS CloudTrail and CloudWatch. Analyzing this data manually is time-consuming and prone to human error, making it difficult to identify potential threats efficiently. This creates a need for an intelligent system that can automatically monitor, analyze, and detect anomalies in real time.

The proposed project focuses on developing an autonomous cyber defense system using AI-based techniques to enhance security in cloud-based ERP environments. The system will continuously monitor cloud logs, identify suspicious patterns, and generate alerts for potential threats. This prototype aims to improve threat visibility, reduce response time, and provide a scalable approach to modern cybersecurity challenges.

Objectives

1. To design and develop an autonomous cyber defense prototype for cloud-based ERP systems.
 2. To implement AI-based techniques for real-time threat detection and monitoring.
 3. To develop a user-friendly dashboard using ASP.NET for visualizing security alerts and logs.
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③ Research Methodology (500 words)

The proposed project is an **application-based system** that focuses on developing a prototype for an autonomous cyber defense mechanism in a cloud-integrated ERP environment. The system is designed to combine cloud monitoring, AI-based analysis, and web-based visualization into a unified architecture.

The methodology begins with the data collection phase, where logs and events are gathered from cloud services such as AWS CloudTrail and CloudWatch. These services capture detailed information about user activities, API calls, and system behavior within the cloud environment. The collected data acts as the primary input for the detection system.

The next phase involves data preprocessing and analysis. In this stage, raw log data is filtered, cleaned, and structured to make it suitable for analysis. AI-based techniques, including basic machine learning models and anomaly detection algorithms, are applied to identify unusual patterns and deviations from normal system behavior. This helps in detecting potential security threats such as unauthorized access, suspicious API calls, and abnormal usage patterns.

The threat detection module processes the analyzed data in real time and generates alerts when suspicious activities are identified. The system simulates an autonomous cyber defense approach by continuously monitoring cloud activity and responding with alerts without manual intervention.

For implementation, the system will be developed using the ASP.NET framework to build the ERP interface and dashboard. The dashboard will provide a user-friendly interface for visualizing logs, alerts, and detected threats in real time. Backend integration ensures seamless communication between cloud services and the detection module.

This project uses **secondary data collection methods**, as it relies on cloud-generated logs rather than direct user input. The overall system is designed as a prototype implementation, focusing on demonstrating key functionalities such as real-time monitoring, automated detection, and alert visualization rather than full-scale deployment.

■ Limitations (200 words)

The proposed system is a prototype implementation and may not fully represent real-world enterprise-level security systems. The AI models used in this project are basic and trained on limited datasets, which may impact the accuracy of threat detection in complex environments. Additionally, the system relies on simulated or sample cloud data rather than large-scale real-time enterprise data.

The project does not include advanced automated response mechanisms such as threat mitigation or system recovery. It focuses only on detection and alert generation. Integration with full-scale enterprise security tools is also beyond the scope of this project.

Performance and scalability may be limited due to resource constraints and simplified architecture. Despite these limitations, the project effectively demonstrates the concept of an autonomous cyber defense system and provides a strong foundation for future enhancements and real-world applications.

🔄 Work Plan (Week 1 to Week 8)

Week No.	Activities Completed
Week 1	a) Study of cloud-based ERP systems and cybersecurity challenges b) Requirement analysis and project scope definition c) Review of existing threat detection techniques
Week 2	a) Designing system architecture b) Identification of modules (data collection, detection, dashboard) c) Selection of tools and technologies

Week 3	a) Setting up AWS environment b) Configuring CloudTrail c) Configuring CloudWatch
Week 4	a) Collecting cloud logs b) Data cleaning and preprocessing c) Structuring data for analysis
Week 5	a) Implementing AI-based detection logic b) Identifying anomalies in logs c) Generating alerts
Week 6	a) Designing ASP.NET dashboard b) Displaying logs and alerts c) Backend integration
Week 7	a) System integration b) Testing functionality c) Bug fixing and optimization
Week 8	a) Documentation preparation b) PPT creation c) Final submission
